



Problem definition

Hateful memes express harm through the combination of image, text, humor, symbols, and cultural references. Their meaning is often **implicit**, so text-only or image-only models may miss the hateful message.



In multilingual settings such as Bangla, detection is harder due to local slang, code-mixing, and community-specific references.

Existing large models can capture richer context, but they are costly to store and deploy, and may still struggle with newly emerging slang, coded symbols, or culture-specific meme references.



Main question: Can knowledge retrieval and teacher supervision improve small multimodal multilingual hate detectors?

We address this by training then fine-tuning a **lightweight MemeBLIP2-based model** using **external knowledge retrieval** and **MemeLens teacher distillation** for efficient Bangla hateful meme detection.

Related works

- Hateful memes require **joint image-text reasoning**, since unimodal models often miss multimodal cues.
- Lightweight models** such as MemeBLIP2 and GatedCLIP improve efficiency by freezing large encoders and training small task-specific modules.
- Context-aware methods** use captions, entities, external knowledge, and hate/offensive lexicons to capture implicit hate.
- Benchmarks such as M3, Multi3Hate, and MUTE highlight the need for models that handle **cross-language and cross-cultural** meme meanings.
- Distillation** transfers knowledge from larger VLMs to smaller models, improving efficiency while preserving stronger contextual supervision.

Our work combines these directions for efficient, knowledge-aware Bangla hateful meme detection.

Architecture



Datasets & Approach

Stage 1: Training on Facebook Hateful Memes Challenge (FHM):

Experiments:

- Learn general hate-meme patterns from a larger English dataset
- Add external knowledge to test whether it can help the model.
- Add (to external knowledge) teacher distillation: test if MemeLens can boost performance.

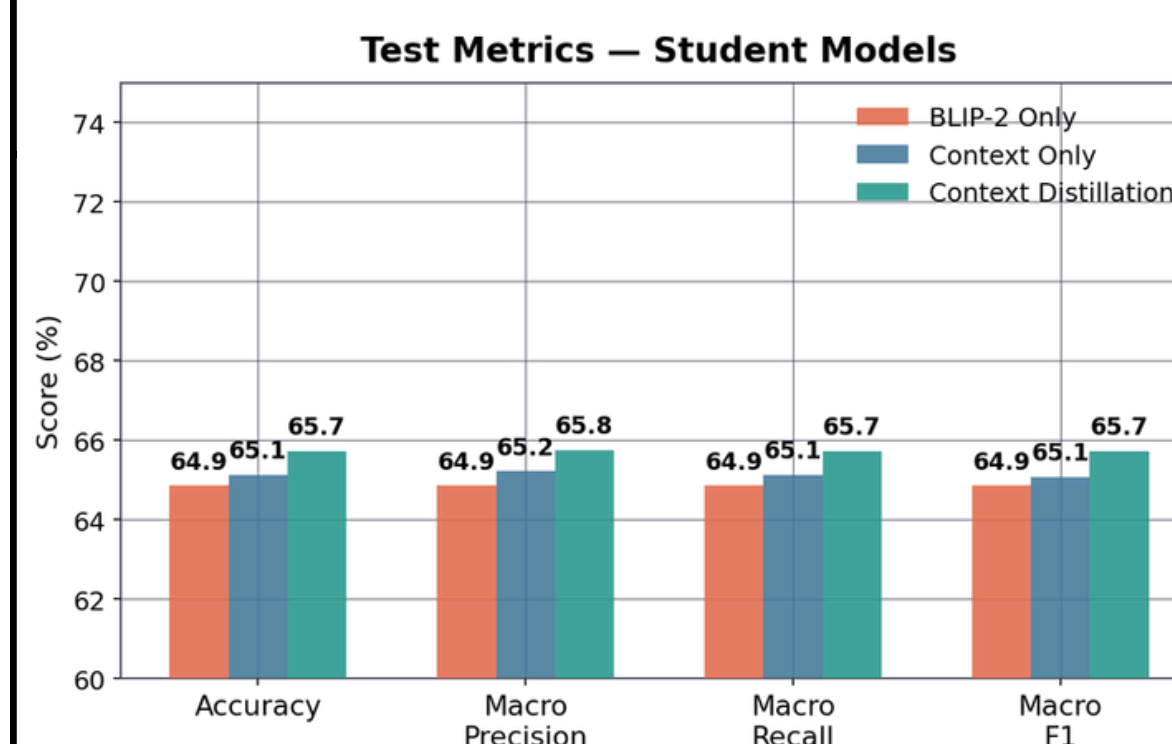
# Hate	# Non-Hate	
3,966	3,966	70% Train 5,552
		15% Val 1,190
		15% Test 1,190

Stage 2: MUTE fine-tuning

- Same 3 experiments, but transfer fine-tune on a smaller Bengali memes dataset, on which MemeLens was also fine-tuned.

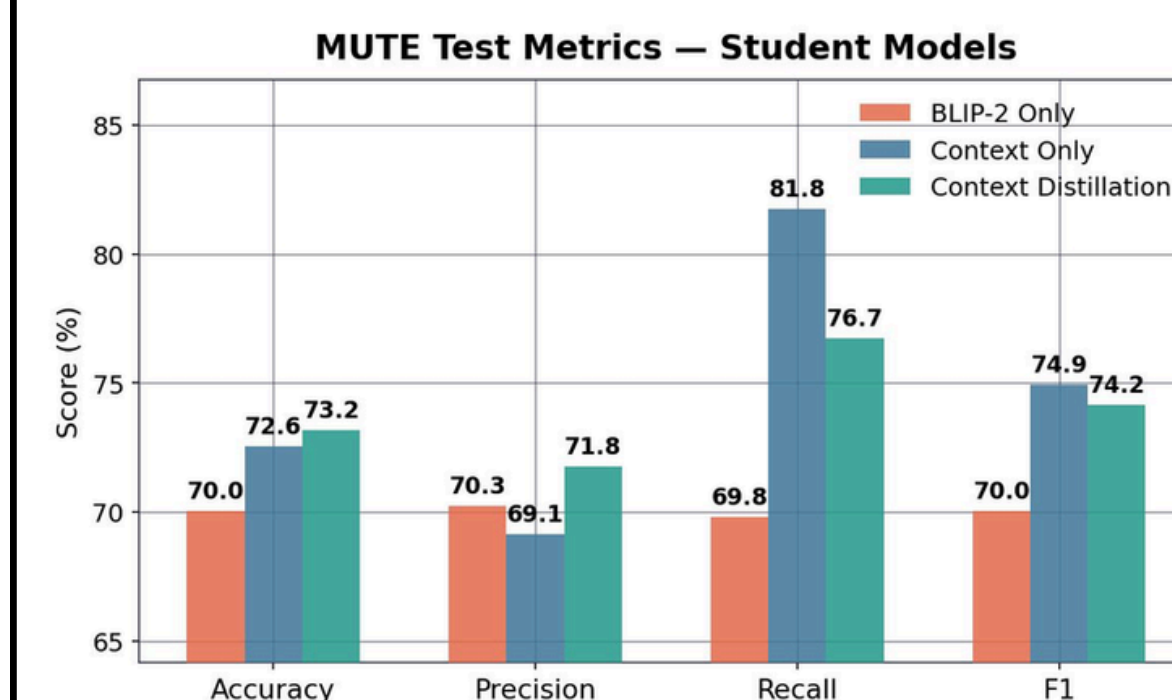
# Hate	# Non-Hate	
1,586	1,586	70% Train 2,222
		15% Val 475
		15% Test 475

Validation



FHM training

- Context adds little:** performance stays close to BLIP-2 only.
- Distillation gives the best score:** but the gain is small.
- Main takeaway:** generic context misses hidden stereotypes.



MUTE fine-tuning

- Context helps strongly:** Bengali hate lexicons gave strong language-specific cues.
- Context + distillation also helps:** improves over BLIP-2
- Main takeaway:** Targeted Bengali hate knowledge outperformed the added distillation signal.

Limitations

- FHM knowledge sources were **too generic and not hate-specific**.
- Retrieved knowledge was linked to text only, **not image regions**.
- Teacher probabilities were parsed from **chat outputs**, not real model logits.
- Small datasets made results **sensitive to overfitting**.

Conclusion

- Lightweight MemeBLIP2-based models can detect hateful memes efficiently, with less storage and inference costs.
- External knowledge helps most when it is language-specific and hate-related: Bengali lexicons improved MUTE results more clearly than generic FHM context.
- Teacher distillation provided only small gains, suggesting limited transfer from chat-based supervision.
- Overall, targeted external knowledge was more useful than generic knowledge or weak distillation.

References

- [1] J. Liu, R. Tong, A. Shen, S. Li, C. Yang, and L. Xu, "MemeBLIP2: A novel lightweight multimodal system to detect harmful memes," arXiv preprint arXiv:2504.21226, 2025. [Online]. Available: <https://arxiv.org/abs/2504.21226>
- [2] Tripathi, S., Kashyap, G. S., Nasim, M., Yang, J., Gao, J., & Naseem, U. (2026). They Said Memes Were Harmless—We Found the Ones That Hurt: Decoding Jokes, Symbols, and Cultural References. In Proceedings of The Web Conference 2026 (WWW 2026). arXiv:2602.03822.
- [3] A. E. Shahroor et al., "MemeLens: Multilingual multitask VLMs for memes," arXiv preprint arXiv:2601.12539, 2026. [Online]. Available: <https://arxiv.org/abs/2601.12539>